

AN FFT SOLVER FOR THE COULOMB PROBLEM

G. LAURITSCH*† and P.-G. REINHARD

Inst. f. Theor. Physik, Universität, D-91058 Erlangen, Germany

Received 22 June 1993

Revised 26 July 1993

A solution of the Coulomb problem in Fourier representation is given. The problems with the long-range parts of the Coulomb potential are circumvented by a separate handling of them. This allows the Fourier representation to develop its typical efficiency and robustness.

Keywords: Poisson Equation; Fourier Representation; 3D Grids; Long Range Separation.

1. Introduction

The electrostatic potential $\Phi(\mathbf{r})$ for a given charge density $\rho(\mathbf{r})$ is given by the Poisson equation

$$\Delta\Phi(\mathbf{r}) = -4\pi e\rho(\mathbf{r}) \quad (1)$$

plus boundary conditions from which a most common condition is

$$\Phi \xrightarrow{|\mathbf{r}| \rightarrow \infty} 0.$$

We thus have a typical boundary condition problem of a second order elliptic differential equation with constant coefficients. The Coulomb problem is ubiquitous in physics and engineering and subsequently there is a vast variety of solution schemes, see e.g., the textbooks.^{1–4} There is no unique answer about the optimal method. This depends on the given problem (e.g., symmetries) and the numerical representation which is already used in other aspects of the problem. Here we want to present a further alternative: a Coulomb solver which works with a coordinate representation of the density ρ and potential Φ and a complementing Fourier-space representation of the Laplace operator Δ . The long range of the Coulomb potential makes a straightforward application to converge badly with the box-size. We take up an idea which has been presented in Ref. 5 already, single out the long-range parts, grant them a separate treatment, and finally recompose the total solution. We call this the FALR method (Fourier Analysis with Long Range forces). The

*Present address: Siemens AG, Medical Engineering, 91050 Erlangen

†Supported by the Deutsche Forschungsgemeinschaft

method is advantageous wherever a Fourier representation is adapted, e.g., on large equidistant grids in full three dimensions.⁶ It shares the efficiency and robustness of the Fourier techniques.

The paper is outlined as follows: In Sec. 2, we present the method; in Sec. 3, we discuss metal clusters as a testcase and compare the FALR method with the SOR (Successive OverRelaxed iteration) scheme which is widely used; and in Sec. 4 we give a short conclusion.

2. The FALR Method

2.1. The problem

The solution of the Poisson equation (1) looks very simple in Fourier representation. We introduce the Fourier transform, $\rho \rightarrow \tilde{\rho}$, for the density

$$\tilde{\rho}(\mathbf{k}) = \mathcal{F}\{\rho(\mathbf{r})\} = \int_{-\infty}^{+\infty} \frac{d^3r}{(2\pi)^{3/2}} e^{i\mathbf{k}\mathbf{r}} \rho(\mathbf{r}) \quad (2)$$

and similarly, $\Phi \rightarrow \tilde{\Phi}$, for the Coulomb potential. The back-transformation looks the same except for the change $+i \rightarrow -i$ in the exponential. The transformation of Eq. (1) then becomes

$$-\mathbf{k}^2 \tilde{\Phi} = -4\pi e \tilde{\rho}$$

which is simply solved as a c-number equation to yield

$$\tilde{\Phi} = \frac{4\pi e}{\mathbf{k}^2} \tilde{\rho} \quad (3)$$

This simplicity is deceiving. It hides two problems: First, the “solver” $1/\mathbf{k}^2$ has a singularity at $k \rightarrow 0$ which can hardly be represented on any finite mesh in $\mathbf{k}$ -space. And second, the unavoidable discretization in coordinate- and Fourier-space turns the Fourier-integrals, see Eq. (2), into a finite Fourier series which subsequently produces a periodic continuation of the density and potential such that

$$\rho(\mathbf{x}_i + n_x L_x, y_j + n_y L_y, z_l + n_z L_z) = \rho(\mathbf{x}_i, y_j, z_l) \quad \text{for all } n_x, n_y, n_z \in \mathbb{Z}$$

where $\mathbf{x}_i, y_j, z_l$ are the mesh points in coordinate-space and L_x, L_y, L_z the grid box-size in each direction. The solution (3) of the Poisson equation in Fourier-space thus implies periodic boundary conditions. The genuinely given isolated charge density is tacitly copied into all periodic equivalent places. Each copy generates its Coulomb field. And finally, the long range of the Coulomb field produces an unphysical interaction between the original density and its copies. This is the well known problem of aliasing, see e.g., chapter 12.1 of Ref. 1.

2.2. Separation of long-range parts

The solution to the above outlined problem is to single-out the dangerous long-range parts of the potential and to treat it separately.⁵ The range of the potential

is related to the multipole component. For large distances, the Coulomb potential becomes⁷

$$\Phi(\mathbf{r}) = \sum_{lm} \frac{4\pi}{2l+1} \frac{Y_{lm}(\mathbf{e}_\mathbf{r})}{r^{l+1}} q_{lm} \quad (4)$$

$$q_{lm} = \int d^3r r^l Y_{lm}(\mathbf{e}_\mathbf{r}) \rho(\mathbf{r}) \quad (5)$$

where Y_{lm} are the spherical harmonics, $\mathbf{e}_\mathbf{r}$ is the unit vector in $\mathbf{r}$ -direction, and the q_{lm} are the spherical multipole moments. The sum runs over $l \in [0, 1, \dots, \infty)$ and $m \in [-l, -l+1, \dots, +l]$. It is obvious that the Coulomb potential of the l th moment behaves as $1/r^{l+1}$ and the long-range parts are thus concentrated in the low- l components. These need to be separated out.

To this end, we introduce a long-range density $\rho^{(\text{long})}$ which takes up all low multipole momenta and which can be solved separately with high precision. This long range density is decomposed in spherical harmonics as

$$\rho^{(\text{long})} = \sum_{lm}^{l_{\text{cut}}} a_{lm} \rho_l(|\mathbf{r}|) Y_{lm}(\mathbf{e}_\mathbf{r}) \quad (6)$$

where ρ_l is some radial reference-density which is arbitrary in principle but in practice should be loosely close to the l th moment of the given density ρ . The choice is not very critical. We will present our particular choice in Subsec. 2.4. The expansion coefficients a_{lm} are to be chosen such that

$$q_{lm}^{(\text{long})} = q_{lm} \quad \text{for } l \leq l_{\text{cut}} \quad (7)$$

where $q_{lm}^{(\text{long})} = a_{lm} 4\pi \int dr r^{2+l} \rho_l(r)$ are the multipole momenta of the long-range density and q_{lm} the momenta of the given density ρ . The expansion coefficients of $\rho^{(\text{long})}$ thus become

$$a_{lm} = \frac{q_{lm}}{4\pi \int dr r^{2+l} \rho_l(r)}. \quad (8)$$

The solution of the Coulomb problem (1) with $\rho^{(\text{long})}$ on the right-hand-side reduces to a one-dimensional radial problem due to rotational invariance of the Laplace operator Δ . We solve the radial Laplace equation

$$\left(\frac{\partial^2}{\partial r^2} + 2 \frac{\partial}{\partial r} - \frac{l(l+1)}{r^2} \right) \Phi_l(|\mathbf{r}|) = -4\pi e^2 \rho_l(|\mathbf{r}|) \quad (9)$$

for each l -component separately. This is effectively a one-dimensional equation and can be solved with arbitrary precision on a fine mesh in radial direction. There are various methods at hand for this task. In practice, we solve the radial Laplace equation (9) by radial Fourier-Bessel forth- and back-transformation. At the end, we compose the radial solutions Φ_l to a solution of the long-range part as

$$\Phi^{(\text{long})}(\mathbf{r}) = \sum_{lm}^{l_{\text{cut}}} a_{lm} \Phi_l(|\mathbf{r}|) Y_{lm}(\mathbf{e}_\mathbf{r}) \quad (10)$$

employing the same expansion coefficients a_{lm} as used in the long-range density (6) and given in Eq. (8).

Having solved the long-range problem with a reference density $\rho^{(\text{long})}$ yielding the corresponding reference potential $\Phi^{(\text{long})}$ we separate the given density ρ into the long-range density $\rho^{(\text{long})}$ and a short-range density $\rho^{(\text{short})}$ as

$$\rho = \rho^{(\text{long})} + \rho^{(\text{short})}. \quad (11)$$

It is clear from Eq. (7) that the short-range density $\rho^{(\text{short})}$ is free of low multipole moments, i.e.,

$$0 = q_{lm}^{(\text{short})} = \int d^3r r^l Y_{lm}(\mathbf{e}_r) \rho^{(\text{short})}(\mathbf{r}) \quad \text{for } l \leq l_{\text{cut}} \quad (12)$$

where l_{cut} is the maximal removed angular momentum. The range in coordinate space is related to the behaviour at $k \rightarrow 0$ in Fourier space. Solution of the Coulomb problem in Fourier space is the simple ratio $\tilde{\rho}/k^2$. Thus $\tilde{\rho}^{(\text{short})}$ should converge fast enough at $k \rightarrow 0$ as to avoid a singularity through $1/k^2$. It is to be noted that

$$\tilde{\rho}^{(\text{short})} \xrightarrow{k \rightarrow 0} k^2 \quad \text{for } l_{\text{cut}} \geq 2.$$

It is obvious that a component with $l = 0$ or 1 would not eliminate the singularity. The components $l = 2$ are not singular but still produce ambiguities at $\mathbf{k} \rightarrow 0$. Thus the solution $\tilde{\rho}/k^2$ is well defined only for

$$l_{\text{cut}} \geq 2 \quad (13)$$

For a safe procedure, one needs to meet the criterion (13). Storage bounds in full three-dimensional calculations, on the other hand, limit the l_{cut} from above such that $l_{\text{cut}} = 2$ is the most practical choice in three-dimensional calculations. This provides an asymptotics $\rho^{(\text{short})} \propto 1/r^5$ which usually gives sufficient convergence with box size.

The solution for the short-range part $\rho^{(\text{short})}$ can now be done with the standard Fourier-space technique

$$\begin{aligned} \rho^{(\text{short})}(\mathbf{r}) &\xrightarrow{\mathcal{F}} \tilde{\rho}^{(\text{short})}(\mathbf{k}) \\ \tilde{\Phi}^{(\text{short})}(\mathbf{k}) &= \frac{4\pi e}{k^2} \tilde{\rho}^{(\text{short})}(\mathbf{k}) \\ \tilde{\Phi}^{(\text{short})}(\mathbf{k}) &\xrightarrow{\mathcal{F}^{-1}} \Phi^{(\text{short})}(\mathbf{r}) \end{aligned} \quad (14)$$

yielding the short-range solution $\Phi^{(\text{short})}$ from $\rho^{(\text{short})}$.

As the last step, we recombine the two potentials to the full solution

$$\Phi(\mathbf{r}) = \Phi^{(\text{long})}(\mathbf{r}) + \Phi^{(\text{short})}(\mathbf{r}) \quad (15)$$

The $Y_{lm}(\mathbf{e}_r)$ in $\Phi^{(\text{long})}(\mathbf{r})$ is given analytically. The radial components $\Phi(|\mathbf{r}|)$ are interpolated from the solutions on the r -space grid, e.g., by cubic splines.

It is to be noted that the scheme works with any reference density $\rho^{(\text{long})}$ whose radial multipole components ρ_l are merely superficially related to the angular projected given density ρ . As we usually know what we expect from physical intuition, we can choose a reasonable set of ρ_l once at the beginning of the procedure, solve the long-range problem only once and use it over and over again in the iterative steps of the whole procedure. This is a key ingredient to the efficiency of the scheme.

2.3. Summary of the procedure

The FALR scheme can be summarized quickly. The *initialization* phase requires five steps:

1. Choose ρ_l .
2. Compute and store $\rho_l(|\mathbf{r}|)Y_{lm}(\mathbf{e}_{\mathbf{r}})$ on a grid in a three-dimensional x, y, z -space.
3. Compute and store the moments $\int dr r^{2+l} \rho_l$ which are required for Eq. (8).
4. Compute the radial solutions $\Phi_l(|\mathbf{r}|)$ on a grid in r -space according to Eq. (9).
5. Compute and store $\Phi_l(|\mathbf{r}|)Y_{lm}(\mathbf{e}_{\mathbf{r}})$ on a grid in three-dimensional x, y, z -space using, e.g., cubic spline interpolation.

The *application* phase requires four steps:

1. Compute q_{lm} for $l \leq l_{\text{cut}}$ and corresponding $m \in \{-l, \dots, +l\}$ according to Eq. (5). The a_{lm} of Eq. (8) follows immediately using the stored $\int dr r^{2+l} \rho_l$.
2. Compose $\rho^{(\text{long})}$ with Eq. (6) and subtract from ρ to obtain $\rho^{(\text{short})}$ according to Eq. (11).
3. Solve the Coulomb problem for $\rho^{(\text{short})}$ with Eqs. (14) yielding $\Phi^{(\text{short})}$.
4. Compose the final $\Phi = \Phi^{(\text{short})} + \sum_{lm} a_{lm} \Phi_l Y_{lm}$ from the $\Phi^{(\text{short})}$ of step 3, a_{lm} of step 1, and from the initially stored $\Phi_l Y_{lm}$.

The expense in CPU time is dominated by the cost of the Fourier forth- and back-transformation in step 3 above. In addition, there are comparatively small costs of computing q_{lm} (one three-dimensional integration) and the various decompositions and compositions in steps 2 and 4. The expense in storage consists in two three-dimensional fields for each lm (for $\rho_l Y_{lm}$ and $\Phi_l Y_{lm}$). This grows quickly with rising l . Thus, one tries to live as long as possible with the minimum required $l_{\text{cut}} = 2$. If one really needs to go much further in l_{cut} , one can save in storage by storing only the radial forms $\rho_l(r)$ and $\Phi_l(r)$ and postponing the interpolation to the x, y, z -grid to each of the later applications. This shifts, of course, expense from storage to CPU time. The method can also be applied to comparable cases in axially symmetric

two-dimensional calculations.¹⁰ In that case, there is little problem with growing storage expense, because there is only one $m = 0$ for each l .

2.4. Choice of reference density

The above scheme is very flexible in formally accepting any desirable reference densities $\rho_l(\mathbf{r})$ for the long-range part. It is obvious that it should be somewhat close to the given density $\rho(\mathbf{r})$. The closest approach would be, of course, the l -projected part $\int d^2\mathbf{e}_r Y_{lm}^*(\mathbf{e}_r)\rho(\mathbf{r})$. But that would require a dynamical update of ρ_l in each step of a surrounding procedure which is supposed to call the Coulomb solver many times with only slightly varying $\rho(\mathbf{r})$. This update requires the above l -projection and the solution of the radial Laplace equation (9) each time anew. But experience shows that a superficial “closeness” of ρ_l to $\rho(\mathbf{r})$ suffices to deliver high precision results on a rather coarse grid. This allows to perform the long-range solution $\rho_l \rightarrow \Phi_l$ only once in the initialization phase. The actual price in each later iteration step is only the little cost of computing the a_{lm} of Eq. (8) and superposing the $\Phi^{(\text{long})}$ according to Eq. (10). This makes the scheme very attractive in all cases where the Coulomb problem is embedded in a larger iterative solution, as e.g., nuclear Hartree-Fock calculations⁸ or Kohn-Sham calculations in electronic systems.⁹

The actual choice in the calculations of the next section has been

$$\begin{aligned}\rho_l &= \frac{2r^l}{R^l + r^l} \rho_0(r) \\ \rho_0(r) &= \left(1 + e^{(r-R)/a}\right)^{-1}\end{aligned}\tag{16}$$

where R is a parameter for the overall radius of the system and a for the expected surface thickness. The choice for ρ_0 is by experience close to the density distribution of most saturating many-Fermion systems (nuclei, metal clusters, liquid ^3He) and has successfully stood the test for those systems. The factor $2r^l/(R^l + r^l)$ guarantees the proper asymptotics $\propto r^l$ for the higher radial moments of the density.

3. Results and Discussion

We take as the testcase a typical charge distribution of a neutral metal cluster Na_{12} as computed in Ref. 9. For an illustration, see Fig. 1 which shows the density ρ_{el} of the valence electrons, the density ρ_J of the positive jellium background, and the total charge density $\rho_{\text{tot}} = \rho_{\text{el}} + \rho_J$ having zero net charge. The example thus readily delivers two very different cases: from ρ_{tot} we obtain a system without monopole moment, i.e., $q_{00} = 0$, and from ρ_{el} a system with charge, i.e., $q_{00} \neq 0$. Both densities display sizeable spatial fluctuations and are thus demanding test cases. We pick the densities from an early stage of the iterative solutions of the Kohn-Sham equations and keep them fixed for the purpose of the present tests. Note that Na_{12} has a strong triaxial deformation with the dimensionless quadrupole momentum $\beta = 0.54$ and triaxiality $\gamma = 15^\circ$, for details see Ref. 9. It also has

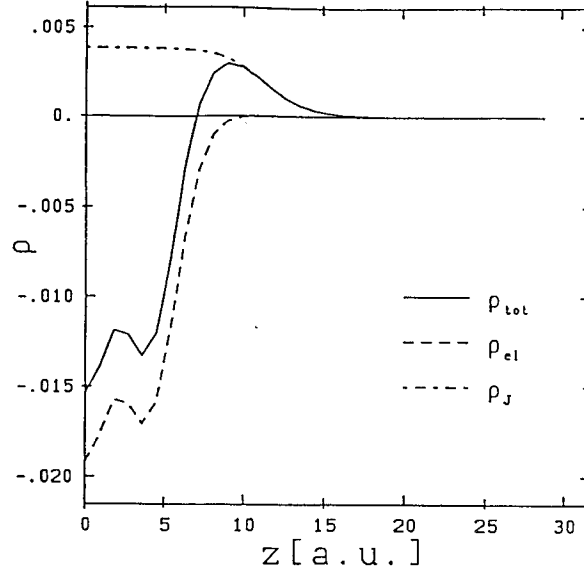


Fig. 1. Charge densities along the z -axis from the early stages of a 3D calculation of the triaxial Na_{12} cluster. $\rho_{tot} \equiv$ total density (full), $\rho_{el} \equiv$ electron density (dashed), $\rho_J \equiv$ jellium background density (dash-dotted).

appreciable higher multipole moments. It is thus a most critical testcase. As the test-observable, we take the Coulomb energy

$$E_{coul} = \frac{1}{2} \int d^3r \rho(\mathbf{r}) \Phi(\mathbf{r})$$

obtained from the given ρ and its corresponding Coulomb field Φ . We measure energies in $Ry = 13.6$ eV and distances in atomic units (a.u.) which are given by $1 \text{ a.u.} = 0.53 \cdot 10^{-10} \text{ m}$. The radius of Na_{12} is about 9 a.u. The oscillations in the electron density have a wavelength of about 4 a.u. The densities are given on an equidistant grid in x, y, z -space with a spacing of $\Delta x = \Delta y = \Delta z = 0.9$ a.u. and $64 \times 64 \times 64$ points. Parity in all three axes allows to reduce the storage to the part $x, y, z > 0$ having $32 \times 32 \times 32$ points. It also grants us the great simplification that all multipole moments q_{lm} vanish for odd l .

We compare the FALR method with a typical conventional Coulomb solver which uses a representation on an equidistant three-dimensional grid together with a three-point formula for the second derivatives in the Laplace operator $\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$, i.e.,

$$\frac{\partial^2}{\partial x^2} f(x, y, z) = \frac{f(x + \Delta x, y, z) - 2f(x, y, z) + f(x - \Delta x, y, z)}{\Delta x^2} \quad (17)$$

and similarly for y - and z -direction. This is one of the most widely used representations. Actually we solve the three-dimensional, discretized Poisson equation with

the SOR method (Successive OverRelaxed iteration). The boundary conditions at the surface of the numerical box are computed from a multipole expansion of the Coulomb potential, i.e., $\Phi = \sum q_{lm} Y_{lm}/r^{l+1}$ with the q_{lm} as given in Eq. 5. In practice, we need the expansion up to $l = 4$ in that case. There are certainly faster iteration schemes, as e.g., the “Chebyshev accelerated successive line overrelaxation” or the ADI method. But here we want to test predominantly the precision and stability of the numerical representation. Thus, speed is not an issue, and we recur to the easily programmable SOR scheme.

The solution of the radial Laplace equation (9) for the initial computation of the Φ_l is rather uncritical. The reference density (16) is much smoother than the given ρ . We find good convergence of the Coulomb energy for an equidistant radial grid with $\Delta r \approx 1.6$ a.u. and $N_r = 35$ grid points. This provides a fast and reliable calculation of the Φ_l in the initialization (although speed is not really critical in that stage of the scheme).

The most interesting feature is the convergence of the solution with the box-size $x_{\max} = N_x \Delta x$ and similarly $y_{\max}$ and $z_{\max}$ because this is very sensitive to the range of the potential. To this end we have fixed the grid-spacing at 0.9 a.u. and varied the number of grid points $N_x = N_y = N_z$. First we show in Fig. 2 the convergence of the Coulomb energy with increasing box-size $x_{\max}$ for the finite-difference scheme (here particularly SOR) with three-point precision according to Eq. (17). Remember that the boundary condition was produced from a multipole expansion up to $l_{\max} = 4$ to achieve sufficient convergence with box-size. The results in Fig. 2 show that the Coulomb energy is very robust (far below 0.1% error) down to boxes of $x_{\max} \approx 20$ a.u. (note the much stretched energy scale). The total energy converges rapidly towards the final result whereas the electron energy improves only slowly in the last digits of the result.

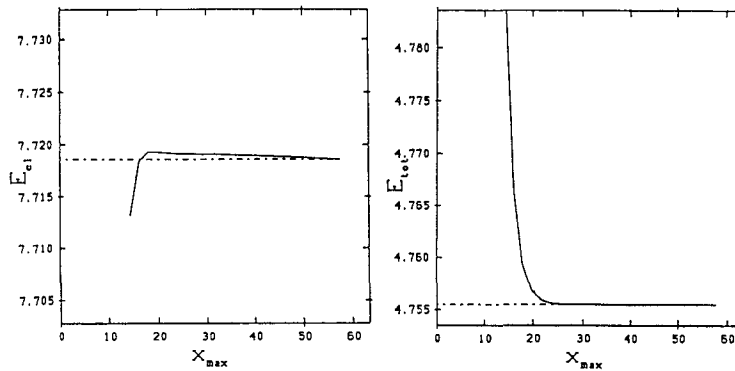


Fig. 2. Electrostatic energy for the solution with the three-point precision (17) for the Laplace operator as function of the box-size $x_{\max}$ for a constant grid spacing of 0.9 a.u. The left part is the energy of the electron density and the right part of the total density. The dashed lines show the energies at the largest boxes of $x_{\max} = 57.6$.

The comparable results for the FALR method with $l_{\text{cut}} = 2$ are given in Fig. 3. We see again robust results in both cases down to fairly small boxes. The electron density, however, converges much slower with increasing box-size. This can be understood from the picture of periodically copied densities, the so called aliases. The potential $\Phi^{(\text{short})}$ behaves as $1/r^5$ for $l_{\text{cut}} = 2$. The interaction with the monopole moment of ρ then produces this slow convergence $\propto 1/r^5$. But the total charge density has vanishing monopole moment $q_{00} = 0$ which enhances the convergence by two orders to $1/r^7$. Thus, $l_{\text{cut}} = 2$ certainly suffices for charge-neutral systems like neutral metal clusters. An application to nuclear fission, however, with its large charges and huge deformations may require higher l_{cut} , or large boxes. Comparison with the finite-difference scheme in three-point precision, see Fig. 2, shows that both schemes behave similar what the convergence with boxsize is concerned. The slightly inferior result for the electron energy comes from the fact that $l_{\text{cut}} = 2 < l_{\text{max}} = 4$. The total energy is well converging in both cases. Note, however, that the converged results for E_{el} and E_{tot} from the both schemes are different. The reason is the different quality of the numerical representation as discussed in the following paragraph.

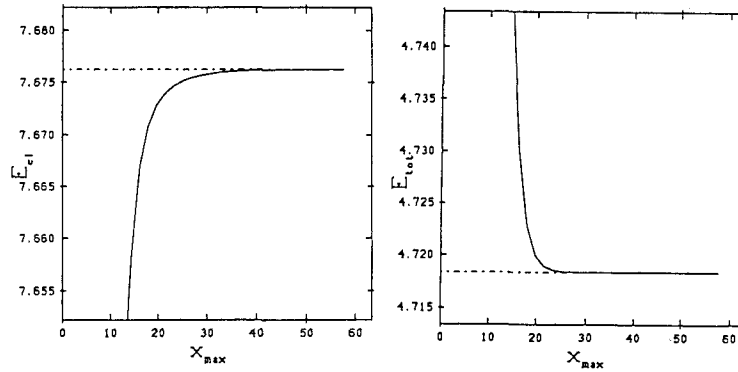


Fig. 3. Electrostatic energy for the solution with the FALR scheme as function of the box-size x_{max} for a constant grid spacing of 0.9 a.u. The left part is the energy of the electron density and the right part of the total density. The dashed lines show the energies at the largest boxes of $x_{\text{max}} = 57.6$.

The schemes behave very differently if we consider the convergence with respect to finer grid-spacings. In Fig. 4, we compare the FALR method with the finite-difference scheme (17). The fixed box-size $x_{\text{max}} = 32$ a.u. is chosen sufficiently large such that the electron energy as well as the total energy have converged. One sees that the FALR method provides a much higher precision and, moreover, behaves very robust up to rather coarse grid spacings. The finite-difference scheme, on the other hand, is still fairly precise (note the much stretched energy scale). But it converges slowly with decreasing grid spacing and never reaches the high precision of the FALR scheme. Attempts to come closer to the final answer by

further reducing the grid spacing in the finite-difference scheme run into difficulties with storage, even on the large CRAY where we did our tests. We could compute finer spacings only while reducing the box-size. But this introduces an error from a too small box and spoils the results far before convergence is reached, see the stars in Fig. 4. Thus, we see that the finite-difference scheme may give a quick calculation with a fair result. But the FALR is much superior in precision and stability. A detailed comparison of the actual speed has not been done for the present example of a Coulomb solver. For that we refer to the extensive comparison of various gridding techniques in Ref. 6 whose findings concerning speed may be taken over for the case considered here.

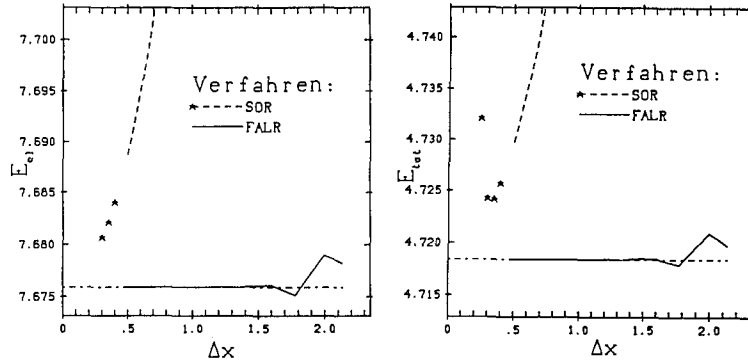


Fig. 4. Electrostatic energy for the solution with the FALR scheme (full lines) and with the finite-difference scheme (dashed lines, denoted SOR) as function of the spacing $\Delta x = \Delta y = \Delta z$ for a constant box size $x_{\max} = 32$ a.u. The left part is the energy of the electron density and the right part of the total density. The stars indicate points with the finite-difference scheme where the box-size $x_{\max}$ had to be reduced in order to make the computation feasible.

4. Conclusions

The Fourier representation of the Laplace operator provides high numerical precision and stability. It becomes particularly efficient in connection with the fast Fourier transformation techniques. But a straightforward application to the Coulomb problem is inhibited, because the long range Coulomb potential interferes with the spurious periodic copies of the charge density which are caused by discretized Fourier series. The FALR method (Fourier Analysis with Long Range forces) solves the problem by subtracting the long-range part and treating it separately. The remaining density is of short range and can be handled with Fourier techniques in a standard manner. The FALR scheme requires only a rough adjustment of the subtracted reference density. It needs to reproduce the first few multipole moments, $l \in \{0, \dots, l_{\text{cut}}\}$ exactly, but its radial shape needs only to be superficially close to actual density, e.g., by adjusting an appropriate global extension. Thus, one needs

to solve the long-range Coulomb problem for the reference density only once in the initialization phase and can then reuse the fixed reference density and its corresponding Coulomb potential arbitrarily often. This makes the method particularly useful for iterative solutions of static and dynamic Hartree-Fock- or Kohn-Sham-equations in saturating many-Fermion systems (as e.g., nuclei, clusters, liquid ^3He) where a rough guess of the spatial extension is provided by the actual Fermion number and the density of the infinite system. The method has already been successfully applied for triaxially deformed metal clusters and nuclear fission.

The FALR method has been tested here for the example of the strongly deformed triaxial metal cluster Na_{12} . This allows to consider two different cases at once: a charge density with monopole moment (for the electrons) and one with vanishing monopole moment (the total density). The shell fluctuations in the density carry high Fourier components and provide a critical and realistic test. A widely used standard technique, finite differences with three-point precision, has been used as normalization point for comparison. We find that the separation of the long-range parts provides indeed a good convergence of the results with increasing box-size. The charge-neutral density is, of course, much better behaved because of the reduced overlap with the residual long-range potential. The charged density is more critical. But it stands the test. For highly charged and more deformed systems, one may have to use larger values than the minimum $l_{\text{cut}} = 2$ which we have employed in the present test. Altogether, the FALR method succeeds in getting the long-range problem out of the way and thus unfolds all the advantages of the Fourier representation which are precision, robustness and efficiency.

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